SIYAMTHANDA AMAHLE BUTHELEZI

DATA ANALYST ● JOHANNESBURG, GAUTENG, SOUTH AFRICA ● 0730203572

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PROFILE

Detail-oriented Data Analyst with over 1 year of experience with a strong foundation in CRM management, business operations, and predictive modeling. Proficient in end-to-end data lifecycle, from Salesforce & SQL extraction and cleaning to deploying machine learning models in Python. Adept at translating complex datasets into clear performance metrics and commercial insights using statistical analysis and visualization tools like Power BI and Seaborn. Proven ability to identify system inefficiencies, implement automation, and apply time series forecasting to drive strategic decision-making.

KEY SKILLS

- Programming & Analysis: Python (Pandas for ETL & Data Cleaning, XlsxWriter for Reporting, NumPy, Scikit-Learn, Statsmodels, Matplotlib, Seaborn), R, SQL, PostgreSQL, Firebase, Advanced Excel.
- Data Science & ML: Time Series Forecasting (Holt-Winters), Classification (Random Forest, Logistic Regression), Regression Analysis, Feature Engineering, Unsupervised Learning (K-Means, PCA).
- Visualization & BI: Power BI, Tableau, Data Storytelling.
- Business & Ops: Salesforce Administration, CRM Management, Power Automate, Process Automation, API Integration, Google Workspace, SharePoint.
- Web & Dev: JavaScript, HTML, CSS, React, Git & GitHub Pages.
- Soft Skills: Stakeholder Management, Technical Documentation, Problem Solving, Agile Methodologies.

KEY COMPETENCIES

• Requirements Elicitation & Documentation • Process Modelling & Optimization • Data Analysis & Visualization • Data Integrity & Quality Assurance • System Design & Architecture • Problem Solving & Critical Thinking • Cross-functional Collaboration • Continuous Learning & Time Management.

EDUCATION

Bachelor of Commerce in Management Information Systems, University of Zululand,
February 2022 — November 2024
Majors: Information Systems and Computer Science

EXPERIENCE

Data & Salesforce Analyst at Medecins Sans Frontieres (Southern Africa),
January 2025 — Present (1 Year 4 Months)

- Data Analysis: Used Python (Pandas) to clean, filter, and analyze marketing data, calculating KPIs such as Customer Acquisition Cost, Win Rates, and Churn.
- Visualization: Built automated reporting dashboards using Matplotlib to visualize revenue trends and conversion funnels for stakeholder presentations.
- Salesforce Data Management & Automation: Developed Python automation scripts (Pandas, XlsxWriter) to process and clean large-scale donor CSV exports, resolving data integrity issues such as character encoding, duplicates, and formatting. Implemented data privacy protocols by automating the masking of sensitive PII (ID/Passport, Credit Card) and auto-generating formatted, weekly call centre batches for operational teams.
- Process Optimization: Engineered an end-to-end workflow automation system using Power Automate and SharePoint to manage internal requests. This solution reduced manual administrative effort and streamlined operations, directly contributing to cost-saving and efficiency improvements.
- Data Governance: Conducted routine health checks on the CRM (Salesforce) to ensure data integrity and quality, reconciling datasets from external sources to maintain compliance with internal standards.

- Reporting & Analysis: Generate weekly and monthly reports on data trends and agent performance using descriptive statistics to summarize key insights, support teams with custom reports and basic analysis, and perform regular health checks on the CRM to ensure data integrity.
- Process Support & Documentation: Document critical Salesforce workflows, data procedures, and call centre data flows to establish internal references and training materials for the team.
- Internal Data Use Mapping: Collaborate on mapping internal data usage by interviewing colleagues to identify gaps in reporting, co-develop a mini-guide on key reports and research best practises for reporting across Digital, Acquisition, Retention, and Philanthropy sectors.
- Testing & Improvements: Test new Salesforce features, automations, or integrations, identify bugs or inefficiencies and propose actionable enhancements to optimize system performance.
- General Support: Assist Data and Retentions teams with additional ad-hoc tasks to support broader organizational goals. Utilized Microsoft Excel (VLOOKUPs, Pivot Tables) to aggregate campaign data.
- Engineered an end-to-end workflow automation system using Power Automate and SharePoint Lists to manage Salesforce and Data requests. Developed logic for role-based routing and automated notifications, delivering a cost-effective internal tooling.
- Presented strategic findings to departmental leadership, shifting the organizational focus from campaign tactics to long-term donor retention strategies.
- Integration Maintenance and error Resolution: Monitor and troubleshoot batch debit order processing between Salesforce and LinkServ via Synatic. Proactively identify system interruptions in the data pipeline and execute corrective actions to resume automated batch processing.

PROJECTS AND VIRTUAL EXPERIENCE

Data Analyst at Quantium/Forage Virtual Programme
March 2026

- Completed a job simulation focused on Data Analytics and Commercial Insights for the data science team.
- Developed expertise in data preparation and customer analytics, utilizing transaction datasets to extract valuable insights and deliver data-driven commercial recommendations.
- Extended analytical capabilities to identify benchmark stores for conducting uplift testing on trial store layouts, enabling evidence-based decision-making.
- Leveraged acquired data analytics and insights from previous tasks to create comprehensive reports for the Category Manager, facilitating informed strategic decisions and enhancing commercial applications.

Data Analyst at Deloitte Australia/Forage Virtual Programme
March 2026

- Completed a Deloitte job simulation involving data analysis and forensic technology.
- Created a data dashboard using Tableau and used Excel to classify data and draw business conclusions.

Front-End Software Engineer at Skyscanner/Forage Virtual Programme
August 2025

- Completed a job simulation where I built a web application using React as a front-end engineer at Skyscanner.
- Developed a page for picking a travel date using Skyscanner's open-source Backpack React library.

Introductory Data Scientist at Commonwealth Bank/Forage Virtual Programme
August 2025

- Completed a job simulation involving Data Management skills for Commonwealth's Data Science Team.
- Demonstrated proficiency in creating data engineering pipelines to aggregate and extract valuable insights from datasets, optimizing data-driven decision-making.
- Acquired skills in anonymizing personal data within datasets, ensuring compliance with data privacy regulations.
- Proposed effective data analysis approaches, particularly related to social media, and demonstrated the ability to design well-structured databases for efficient information management.

Student Fingerprint Verification System at University of Zululand
Capstone Project | June 2024 – November 2024

- Spearheaded requirements documentation (SRS) and system design documentation (SDD) for a biometric student attendance system, translating stakeholder needs into technical specifications.
- Researched and implemented Support Vector Machine (SVM) fingerprint recognition using Python libraries (scikit-learn, OpenCV), achieving exceptional performance metrics:
  - Precision: 93.82%
  - Accuracy: 89%
  - Recall: 83.5%
- Navigated project constraints when institutional budget limitations prevented acquisition of the fingerprint scanner SDK, requiring rapid adaptation of demonstration approach.
- Engineered a simulated environment to showcase system functionality to stakeholders, including university administrators and faculty, demonstrating core features and value proposition.
- Presented system architecture, model performance metrics, and implementation roadmap to a live audience, receiving positive feedback on analytical approach and problem-solving adaptability.
- Documented detailed requirements for full implementation, including SDK specifications and integration requirements, to enable future deployment.

PERSONAL DATA SCIENCE PROJECTS

South Africa Inflation Analysis Trilogy (Python, Pandas, Statsmodels)

- Forecasting: Built a Holt-Winters Exponential Smoothing model to forecast 2025-2026 inflation rates, achieving a Mean Absolute Error (MAE) of 0.43% against economist projections.
- Volatility Analysis: Analysed 50 years of economic data to identify structural breaks and "eras of chaos," calculating rolling volatility to correlate historical events (e.g., 1994 Elections, 2008 Crisis) with market instability.
- Purchasing Power Simulation: Engineered a simulation model to compare nominal wage growth against cumulative inflation, visualizing the "silent erosion" of real wages and the long-term impact of economic trends on consumer purchasing power.

iPhone Resale Market Intelligence (Python, Scikit-Learn, Pandas)

- Predictive Modelling: Developed a Random Forest Classifier to predict "High Demand" listings (71% accuracy) and a Regression model to estimate housing prices, utilizing feature engineering (Price Z-Score, Supply Saturation).
- Market Analysis: Identified "Dead Inventory" configurations (0 sales, high supply) and performed Elasticity Analysis to determine optimal pricing strategies for specific tech models.

Global Homicide Trends Analysis [1970–2030] (Python, Scikit-Learn, Statsmodels)

- Unsupervised Learning (Clustering): Applied K-Means Clustering to 13 nations to segment them into distinct "Crime Trajectories" (Safe Havens vs. Critical Zones), identifying South Africa and Colombia as distinct outliers.
- Time Series Forecasting: Utilized Holt-Winters Exponential Smoothing to project homicide rates for South Africa and Colombia into 2030, predicting stagnation and slight increase respectively.
- Divergence Analysis: Visualized the "Global Safety Gap" over 50 years, revealing persistent inequality between the safest (Japan/Singapore) and most dangerous nations.

Walsoft Institute: Student Success Optimization (Python, Pandas, Scikit-Learn)

- Strategic Analysis: Conducted a full-cycle consulting analysis to optimize student performance and resource allocation for a Business Intelligence training program, analyzing demographic, academic, and behavioral data.
- Data Wrangling: Engineered a robust ETL pipeline to clean a messy raw dataset, resolving inconsistencies in categorical data (Country, Residence, Gender) and handling missing values via imputation.

- Predictive Modelling: Built a Random Forest Regressor to identify performance drivers, uncovering that Study Hours drive 68% of Python success (Effort-based) while Entry Exams drive 60% of DB success (Aptitude-based).
- Business Impact & ROI: Developed a segmentation strategy identifying at-risk groups (e.g., specific residence types) and recommended a "Gamified Study Hour" intervention as the highest-ROI strategy to improve pass rates.
- Deliverables: Automated the generation of executive reports, strategic recommendations, and visualization dashboards (Correlation Heatmaps, Feature Importance plots) for stakeholder review.

RELEVANT COURSEWORK

- Health Management Information Systems, Tembo | March 2026
- Introduction to Python, DataCamp | February 2026
- Intermediate SQL, DataCamp | February 2026
- Introduction to SQL, DataCamp | February 2026
- IT Customer Support Basics, Cisco Networking Academy | May 2025
- Introduction to Cybersecurity, Cisco Networking Academy | April 2025
- JavaScript Essentials 1, Cisco / OpenEDG | January 2025
- SQL and Relational Databases 101, Cognitive Class | March 2024
- Introduction to Packet Tracer, Cisco Networking Academy | March 2023